Tutorial-1

(Dimensional analysis, Order-of-magnitudes, Mathematical analysis)

1. **Dimensional analysis:**

a) Show that the dimensions of the equation are consistent.

$$V_f^2 = V_i^2 + 2\,a\,(x_f - x_i)$$

Where V_f and V_i are final and initial velocities, and x_f and x_i are final and initial position, respectively.

b) If both sides are multiplied by 'm' (a mass), what common dimensions do both sides have and what new derived dimension is this?

2. **Estimation/Order of magnitudes:**

(i). What is your best estimate for the volume of the earth's atmosphere?

(A). $(10^1 - 10^5)m^3$, (B). $(10^5 - 10^{10})m^3$, (C). $(10^{10} - 10^{15})m^3$, (D). $(10^{15} - 10^{20})m^3$.

(Hint: Earth's radius: 6.5 x 10^3 Kms.; Thickness of earth's atmosphere: 10 Kms.)

(ii). SNU campus has ~500 street lamps. How much does each lamp cost to operate per year? Assume each street lamp is 300 Watt and cost of electricity is Rs.2 per unit and street lights are getting "ON" for 8 Hrs a day.

3. **Geometrical analysis:**

Prove that surface area of a sphere (radius r) is $4\pi r^2$. Do the same question but using dimensional analysis.